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Non-Intrusive Load Monitoring (NILM) with very low-frequency data from smart meters in Switzerland

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ABSTRACT

In Switzerland, the residential sector accounts for approximately 34 % of total energy consumption. Understanding the energy consumption patterns of residential customers is crucial for developing effective energy efficiency strategies. Traditional methods for collecting energy consumption data rely on installing smart meters on individual appliances, which can be intrusive and expensive. Non-Intrusive Load Monitoring (NILM) offers a promising alternative by disaggregating the total energy consumption of a household into individual appliance-level energy consumption data. The frequency of data from smart meters can vary depending on the specific type of smart meter and the utility company's data collection practices. In general, smart meters can collect data at a very high frequency, often in the range of minutes or even seconds. However, it is typical for mass-deployed smart meters to collect data at a frequency of 15 min, e.g. in Europe and Switzerland. This paper explores different NILM algorithms suited to very low frequency (1-min to 15-min sampling rate) metered data. Particularly the performance of the different algorithms is trained and tested on (i) same and (ii) different dataset using smart meter data at (i) 1 min, (ii) 5 min and (iii) 15 min frequency. The performance of NILM algorithms is highly dependent on the appliance, sample period, and dataset specificity. SGN emerges as the most versatile algorithm, excelling in capturing both dynamic and steady-state appliance behaviors across different sampling rates and validation scenarios.

1. Introduction

In Switzerland, energy consumption patterns reflect a strong reliance on residential, industry, and commercial sectors. Over the past two decades, energy use in the residential and commercial sectors has steadily increased, underscoring Switzerland's urbanization and high living standards. The above-mentioned trends as reflected in Fig. 1, which illustrates the evolution of final electricity consumption by different sectors in Switzerland between 2000 and 2022. The industrial consumption, on the other hand, has fluctuated, reflecting shifts in industrial activities and energy efficiency improvements. Similarly, in other European countries, residential energy consumption has been driven by space heating, accounting for about 65 % of household energy use. Energy efficiency efforts, such as improved building insulation and energy-efficient appliances, have reduced energy consumption in many areas, though trends vary across regions and climate conditions [1,2]. The transport and agriculture sectors have maintained low and relatively stable energy demands, consistent with Switzerland's robust public transportation network and limited agricultural land area. This aligns with trends observed across Europe, where lifestyle changes, appliance ownership, and renovation of older building stocks have been significant factors influencing energy use. Fig. 2 below illustrates the final electricity consumption by the prominent sectors in Switzerland as recorded in 2022. Focusing on the residential sector, it is a significant consumer of energy, accounting for approximately 34 % of the Switzerland's total energy consumption [3]. And this pattern again mirrors the broader European trend of residential buildings contributing a large portion of final energy consumption.

Much of this energy is used for heating and domestic hot water, followed by electrical appliances and lighting [4]. The electrical appliances used for cooking, refrigeration, laundry, entertainment and IT constitute about 40 % of the total electricity consumption in Swiss buildings as shown in Fig. 3. This significant share highlights the potential for energy efficiency improvements and the adoption of smarter, more energy-efficient appliances to reduce electricity consumption in residential buildings. Switzerland's commitment to reducing its carbon footprint and transitioning to a more sustainable energy system is evident in its efforts to promote energy conservation and the use of

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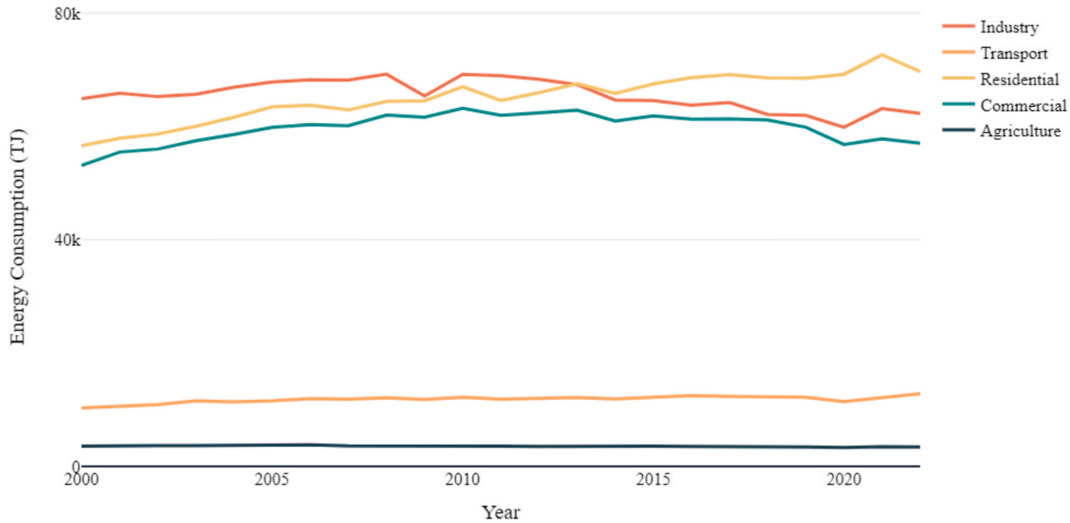

Fig. 1. Evolution of electricity final consumption by sector in Switzerland in the last two decades.

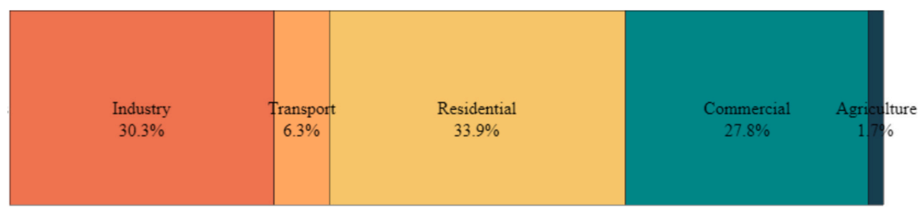

Fig. 2. Electricity final consumption by Sector in Switzerland, 2022.

renewable energy sources within homes [5]. It includes the reduction of energy consumption per capita and the increase in the share of renewable energies in its overall energy mix as outlined in the Energy Strategy 2050 [6]. Switzerland improved its energy efficiency by 1.4 % per year from 2000 to 2016, with household being the fastest sector and industry the slowest, but still all individual sectors need to increase their rate of improvement to meet the Swiss Energy Strategy 2050 targets [7]. Consumers are also willing to pay for energy-saving measures in residential buildings, valuing benefits such as individual savings, environmental benefits, and comfort benefits [8].

Non-Intrusive Load Monitoring (NILM) is the process of estimating the energy consumed by individual appliances given just a higher-level power meter reading without requiring access to the individual consumers and producers [9]. By analysing changes in the voltage, current, frequency and/or power readings going into a home, NILM algorithms can disaggregate the total energy consumption into device-specific usage data, without the need to install additional metering equipment on each appliance [10,11]. The applications of NILM are potentially beneficial for both consumers and energy providers. These techniques can efficiently disaggregate electricity into specific appliance loads, aiding in Home Energy Management Systems and Ambient Assisted Living [12]. For utility companies, NILM facilitates demand-side management, improves load forecasting accuracy, supports the implementation of dynamic pricing models and enhance residential electricity audits by providing detailed, continuous appliance-level data [13]. Additionally, it plays a crucial role in energy research and building operation by providing detailed energy consumption insights. The subject of interest in this work is primarily residential buildings, however commercial buildings also present an interesting challenge in load disaggregation given its load complexity, small changes in power consumption and continuous operating loads [14].

Swiss energy policy and utility companies have been adopting smart metering technologies to increase energy efficiency and integrate renewable energy sources [15]. The increasing deployment of smart meters opens opportunities for NILM to further optimize energy consumption. The latest developments with respect to smart meters in Switzerland focus on the nationwide rollout of advanced metering infrastructure, mandated by Swiss energy legislation. The effectiveness of NILM techniques is significantly dependent on the frequency of data collected by smart meters. High-frequency data allows for more granular and accurate appliance identification, but most residential smart meters record at lower frequencies, typically 15-minute intervals. This poses a challenge for NILM algorithms, which must be sophisticated enough to infer appliance signatures from less detailed data. These challenges include the loss of temporal detail, which hinders device identification and state detection, reduced algorithm performance as most NILM methods excel with high-frequency data but degrade significantly at lower sampling rates, and data quality issues like missing data or gaps that complicate disaggregation [16–20]. To address these issues, recent research has focused on deep learning and model innovations, including the use of deep neural networks such as LSTM and GAN-based architectures to enhance feature extraction and classification from low-frequency data [21,22]. Deep neural networks can effectively disaggregate appliances from low frequency data but require multiple input features and comparative studies for successful deployment [23,24].

Basu et al. proposed a low sampling rate time series distance-based approach for non-intrusive load monitoring in residential buildings and demonstrated its effectiveness in disaggregating individual appliances from total load [25]. Perez et al. proposed a disaggregation technique targeted specially towards air-conditioning energy use. This work also resonated the importance of the data granularity in achieving accurate disaggregation performance [26]. Athanasiadis et al. have established real-time NILM implementations with their lightweight and scalable approach that enables multi-class appliance detection even with the constraints of low-frequency sampling, making it particularly suitable for embedding into resource-constrained smart meter devices [27,28]. Virtsionis et al. have advanced the methodological framework through their suite of deep learning innovations, notably with their

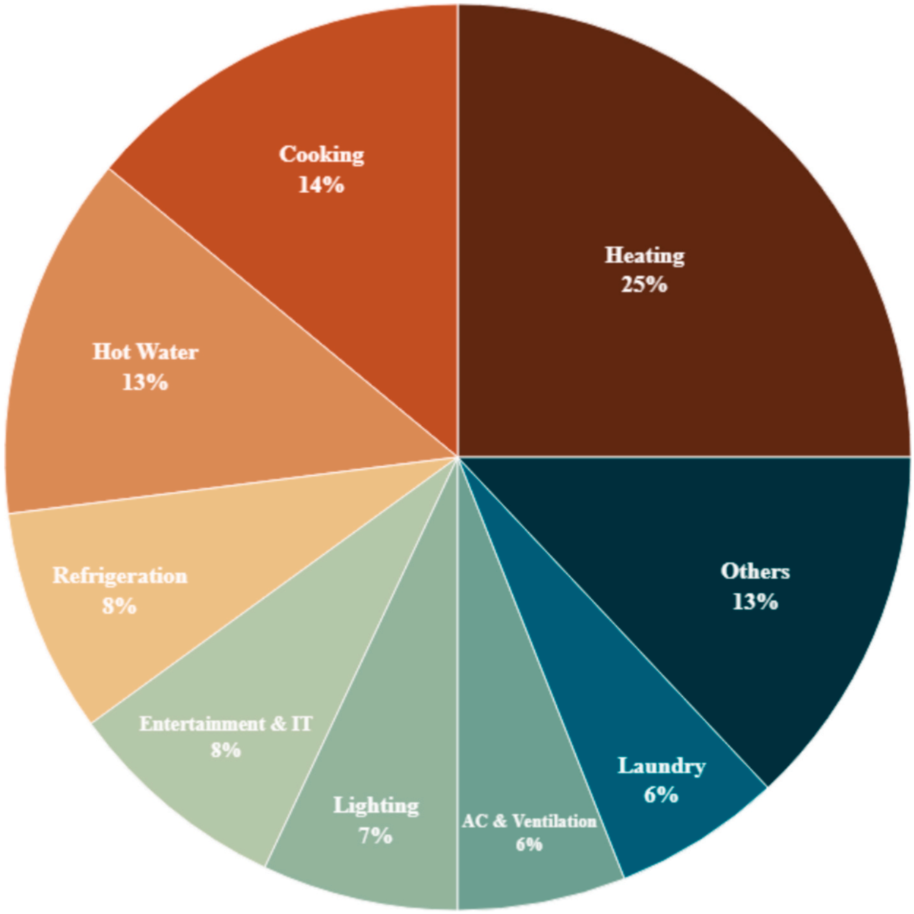

Fig. 3. Electricity use by purpose in typical Swiss residences.

Neural Fourier Energy Disaggregation architecture that employs Fourier transformations to reduce computational parameters while maintaining competitive performance metrics [29] and variational regression model for multi-appliance power disaggregation [30]. Their development of Torch-NILM represents a significant contribution as the first NILM-specific deep learning toolkit built on PyTorch, facilitating standardized experimentation for researchers working with low-frequency data through its comprehensive APIs for benchmarking, hyperparameter tuning, and cross-validation [31].

Based on multiple research, it was also concluded that incorporating feedback mechanisms and multi-task learning further improves model robustness, especially with imbalanced or sparse dataset [16,17,21,22]. Transfer learning and active learning frameworks facilitate model adaptation to new environments with minimal labelled data, enhancing scalability and reducing retraining efforts. Data handling techniques such as augmentation to artificially increase sampling rates or fill data gaps, and compressed sensing principles applied to smart meters, have shown promise in improving appliance state detection from sparse data [18,32]. Algorithmic strategies also include unsupervised and statistical methods like Markov models and K-nearest neighbours, which can disaggregate appliance categories without labelled data, offering practical solutions for 15-minute interval data [19,20]. Feature engineering using statistical features, time-of-use profiles, and reactive power measurements can compensate for the lack of temporal resolution. The development of low-frequency datasets, including the Indian Energy Dataset with Low frequency (IEDL), supports research and application in diverse contexts [33]. Benchmarking studies indicate that while supervised methods generally outperform unsupervised ones, the latter provide advantages in cost and computational efficiency for low-frequency data.

In summary, NILM at very low frequency data remains challenging due to limited temporal information and data quality issues. However, advances in deep learning, transfer learning, data augmentation, and unsupervised methods are progressively improving performance, though trade-offs between accuracy, scalability, and computational demands persist, necessitating further research to bridge the gap with high-frequency NILM solutions. This paper explores different deep-learning based NILM algorithms suited to low frequency metered data (5-min to 15-min sampling rate). The performance of the different deep learning-based algorithms like Denoising Auto Encoder (DAE), Bidirectional Long short-term memory networks (BiLSTM), Semantics-Guided Neural Networks (SGN) and Sequence-to-point learning (Seq2point) are tested. This analysis is performed on REFIT and ECO dataset with the training and testing is done on (i) same and (ii) different dataset. Additionally, different sample periods i.e. 1 min, 5 min and 15 min are considered to analyse the variation in disaggregation accuracy by reducing the data sampling rate. One key feature that sets this research apart is its innovative focus on exploring NILM algorithms tailored to the Swiss context particularly to low frequency smart meter data, which remains an underexplored area in the existing literature. Novel aspects also include cross-dataset validation between different national contexts, comprehensive comparison of classical and deep learning approaches, and investigation of appliance interference patterns on disaggregation accuracy. The cross-dataset validation methodology tests algorithm generalizability between British (REFIT) and Swiss (ECO) datasets provide valuable insights into the transferability of NILM models across different national contexts. These findings advance understanding of NILM algorithm performance in real-world, low-frequency smart meter deployments, with relevance to the Swiss energy landscape.

Table 1
Appliances measured in the REFIT and ECO dataset.

dataset	no.	start date	end date	food related								cleaning			entertainment			utilities	
				fridge	freezer	fridge freezer	coffee maker	kettle	dish washer	micro wave	toaster	washing machine	washer dryer	tumble dryer	PC	TV	audio	router	lamp
eco	1	2012.06.01	2013.01.31	1	1	0	1	1	0	0	0	1	0	0	1	0	0	0	0
	2	2012.06.01	2013.01.31	1	1	0	0	1	1	0	0	0	0	0	0	1	1	0	1
	3	2012.06.27	2013.02.01	1	1	0	1	1	0	0	0	0	0	0	1	0	0	0	0
	4	2012.06.27	2013.02.02	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	1
	5	2012.06.27	2013.02.03	1	0	0	1	1	0	1	0	0	0	0	1	0	0	0	0
	6	2012.06.27	2013.02.04	1	0	0	1	1	0	0	0	0	0	0	0	0	0	1	1
refit	1	2013.10.09	2015.07.10	1	1	0	0	0	1	0	0	1	1	0	1	1	0	0	0
	2	2013.09.17	2015.05.28	0	0	1	0	1	1	1	1	1	0	0	0	1	1	0	0
	3	2013.09.25	2015.06.02	0	1	1	0	1	1	1	1	1	0	1	0	1	0	0	0
	4	2013.10.11	2015.07.07	1	1	1	0	1	0	1	0	1	0	1	0	1	0	0	0
	5	2013.09.26	2015.07.06	0	0	1	0	1	0	1	1	1	0	1	1	1	0	0	0
	6	2013.11.28	2015.06.28	0	1	0	0	1	1	1	1	1	0	0	1	1	0	0	0
	7	2013.11.01	2015.07.08	1	1	0	0	1	1	0	1	1	0	1	0	1	0	0	0
	8	2013.11.01	2015.05.10	1	1	0	0	1	0	1	1	1	1	0	1	1	0	0	0
	9	2013.12.17	2015.07.08	0	0	1	0	1	1	1	0	1	1	0	0	1	1	0	0
	10	2013.11.20	2015.06.30	0	1	1	0	0	1	1	1	1	0	0	0	1	0	0	0
	11	2014.06.03	2015.06.30	1	0	1	0	1	1	1	0	1	0	0	1	0	1	1	0
	12	2014.03.07	2015.07.08	0	0	1	0	1	0	1	1	0	0	0	1	1	0	0	0
	13	2014.01.17	2015.05.31	0	1	0	0	1	1	1	0	1	0	0	0	1	0	1	0
	15	2013.12.17	2015.07.08	0	0	1	0	0	1	1	1	1	0	1	1	1	1	0	0
	16	2014.01.10	2015.07.08	0	0	1	0	0	1	0	0	1	0	0	1	1	0	0	0
	17	2014.03.06	2015.06.19	0	1	1	0	1	0	1	0	1	0	1	1	1	0	0	0
	18	2014.03.07	2015.05.24	1	1	1	0	0	1	1	0	1	1	0	1	1	0	0	0
	19	2014.03.06	2015.06.20	0	0	1	0	1	0	1	1	1	0	0	0	1	1	0	0
	20	2014.03.20	2015.06.23	1	1	0	0	1	1	1	0	1	0	1	1	1	0	0	0
	21	2014.03.07	2015.07.10	0	0	1	0	0	1	0	0	1	0	1	0	1	0	0	0

2. Methodology

In this chapter, the methodology employed to advance the study on NILM algorithms in the Swiss context is described. Initially, the different datasets instrumental in training and evaluating NILM algorithms is explained. These datasets have been selected to encompass a variety of appliances, ranging from high-energy-consuming devices such as washing machine to smaller but more transient loads like freezer. The chapter also presents an explanation of the array of deep-learning algorithms that have been chosen for the purpose of this study. It is followed by an exploration of the NILM frameworks that serve as the foundation for implementing these algorithms, offering a standardized environment for comparison and analysis. To assess the efficacy of the algorithms, a suite of metrics is chosen which is also discussed in this chapter. This comprehensive methodological framework sets the stage for a robust analysis of NILM systems, facilitating an understanding of their practical applications and limitations in real-world energy management scenarios.

In this study two popular open-source datasets, i.e. REFIT and ECO were used. The REFIT dataset [34] contains cleaned electrical usage data measured in Watts from 20 homes over a two-year period. The data includes whole-home energy use and use of individual appliances, recorded every 8 s. This information can be used for research on energy savings, smart technology applications for homes, analysing appliance usage, and more. The data has undergone pre-processing to clean and prepare it for analysis. This includes merging duplicate timestamps, removing unrealistic readings from individual appliances, ensuring each appliance monitor only shows one device, updating notes on appliance changes, and filling in or removing short gaps in data. This data was collected as part of the REFIT project funded by EPSRC to develop personalized retrofit recommendations for UK homes.

ECO dataset [35,36] includes detailed electricity consumption and occupancy data collected over 8 months from six Swiss homes. The dataset contains, (i) Household electricity use measured every second including current, voltage and phase for each phase, (ii) Plug-level power measurements of selected appliances at 1-second intervals and, (iii) Occupancy information through a tablet (manually labelled) and PIR sensor readings for some homes. This data was gathered as part of a project called Smart Meter Services which focused on non-intrusive load monitoring and occupancy sensing using smart meters. The dataset is used to evaluate different load disaggregation algorithms and occupancy detection methods through an open-source framework called NILM-Eval [37]. The Table 1 shows details about both the considered dataset in terms of the timeframe of the measurements and the appliances monitored in each building.

The primary goal of the study was to explore the load disaggregation performance of certain deep learning algorithms with low resolution smart meter data. The algorithms shortlisted for the exploration were DAE, BiLSTM, SGN and Seq2point. DAE has gained prominence for its ability to reconstruct clean signals from corrupted inputs, an attribute that makes it particularly adept at handling the noisy data often encountered in energy disaggregation tasks [38]. BiLSTM leverages its bidirectional information processing to capture temporal dependencies and patterns in both forward and reverse time order, thus providing a holistic view of the energy usage patterns [38]. Contrastingly, SGN derives its strength from its unique approach of extracting distinctive features from the electrical load, which encapsulates the idiosyncratic operational modes of individual appliances [39]. Seq2Point learning

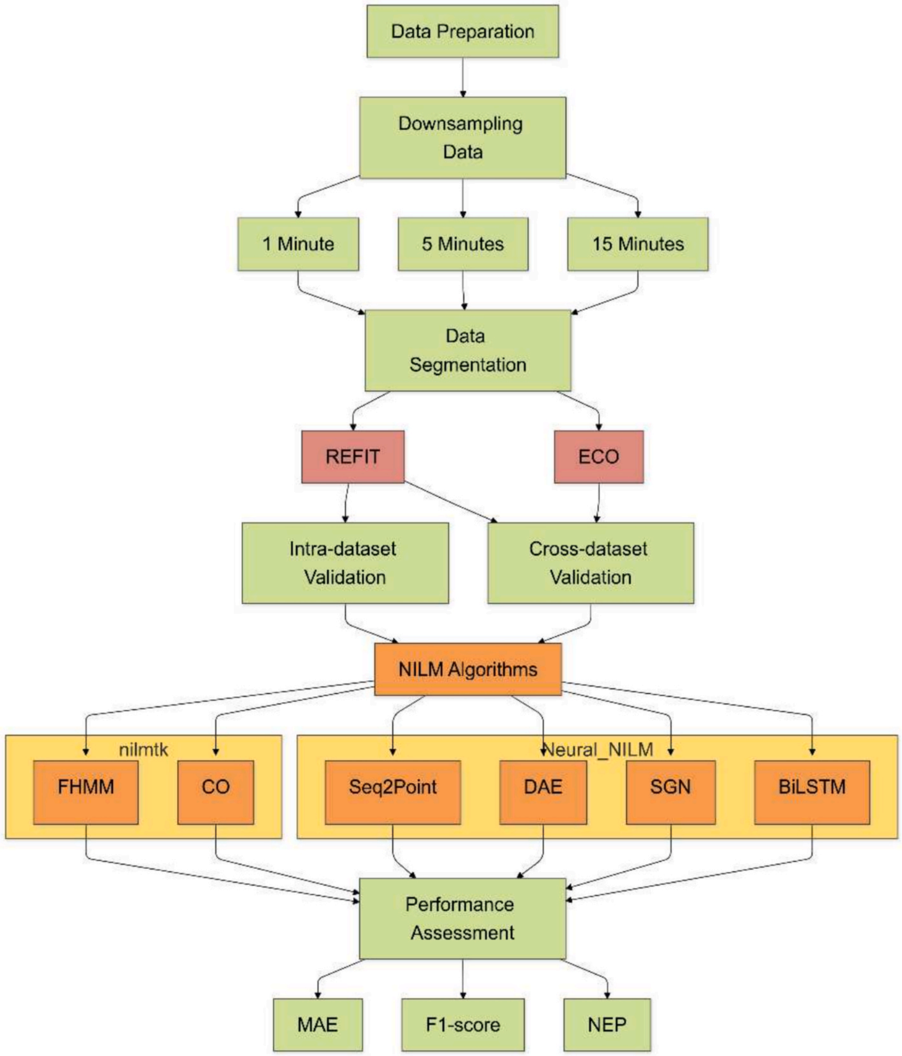

Fig. 4. Workflow of NILM Data Processing and Performance Assessment Framework.

model focuses on predicting the midpoint of an appliance's power consumption sequence and benefits from this narrowed scope to deliver precise disaggregation performance [40]. Each of these models have their own advantages and challenges but they are united by the fact that they all have shown promising results in disaggregating energy data from low resolution smart meters. For comparison purposes, two benchmark NILM algorithms i.e. FHMM (Factorial Hidden Markov Models) and CO (Combinatorial Optimization) were also considered. CO and FHMM are often considered benchmark methods for NILM due to their proven effectiveness, adaptability, and consistent performance in accurately disaggregating household energy consumption data. CO is valued for its simplicity and reliability, while FHMM is recognized for its advanced modelling capabilities and adaptability to real-world conditions [41,42].

In an effort to build upon the established knowledge within the field and to expedite our research, the study leveraged a couple of existing frameworks, nilmtk [43] and NeuralNILM [44]. These frameworks have been tested and refined through continued use in the NILM community. nilmtk is a toolkit designed to help researchers evaluate the accuracy of NILM algorithms. It is an open-source toolkit designed specifically to enable the comparison of energy disaggregation algorithms in a reproducible manner. This work is the first research to compare multiple disaggregation approaches across multiple publicly available data sets. Nilmtk includes, (i) parsers for a range of existing data sets, (ii) collection of pre-processing algorithms, (iii) set of statistics for describing data sets, (iv) number of reference benchmark disaggregation algorithms, and (v) common set of accuracy metrics. The other framework NeuralNILM provide code repositories implementing some state of the art (or classical) energy disaggregation algorithms by the means of prevailing deep learning toolkit PyTorch [45] and nilmtk. In the quantitative analysis of the above-mentioned algorithms the metrics chosen were Mean Absolute Error (MAE), F1 Score and Normalized Error in Assigned Power (NEP). MAE provides a measure of the average magnitude of errors between the predicted and actual appliance power consumption. F1 Score is calculated as a harmonic mean of precision and recall and offers a balanced view of a model's accuracy. Precision is the ratio of correctly predicted positive observations to the total predicted positives and recall is the ability of the model to find all the relevant cases within a dataset. Additionally, NEP provides a more NILM-focused assessment, quantifying the error in energy assigned to each appliance against the total energy consumption. Collectively, the above metrics offer a comprehensive toolkit to assess the strengths and weaknesses of NILM algorithms.

Performance of NILM algorithms was systematically evaluated on two benchmark datasets, REFIT and ECO. The primary steps involved in the NILM analysis performed in this study is illustrated in the flowchart in Fig. 4. The models were subjected to both intra-dataset and cross-dataset validation; they were trained and tested on the same dataset, as well as on differing datasets. Within the REFIT dataset, models were trained using data recorded over a one-year period, from 07/03/2014 to

Table 2
Performance of NILM algorithms for various appliances at different sampling frequencies.

Test	Sample Freq.	Appliance	MAE (W)						F1 Score (-)						NEP (-)					
			FHMM	CO	Seq2 Point	SGN	DAE	BiLSTM	FHMM	CO	Seq2 Point	SGN	DAE	BiLSTM	FHMM	CO	Seq2 Point	SGN	DAE	BiLSTM
REFIT	1 min	kettle	174.0	294.0	15.0	9.1	17.5	21.7	0.15	0.07	0.68	0.77	0.62	0.75	8.39	14.17	0.84	0.51	0.98	1.22
		freezer	48.0	136.0	28.2	25.0	32.1	39.0	0.42	0.43	0.42	0.49	0.40	0.38	2.13	6.04	1.45	1.29	1.65	2.01
		washing machine	302.1	224.0	20.2	14.1	36.3	47.1	0.17	0.06	0.64	0.76	0.30	0.10	7.87	5.83	0.61	0.43	1.10	1.42
		computer	42.2	30.1	4.0	5.0	5.1	9.3	1.00	0.68	1.00	1.00	1.00	0.70	2.80	1.99	0.27	0.33	0.34	0.62
		micro-wave	238.0	448.5	10.8	7.4	12.5	12.7	0.02	0.02	0.21	0.35	0.18	0.11	23.24	43.81	1.19	0.82	1.38	1.40
		television	81.2	76.7	11.8	10.2	12.2	12.7	0.31	0.37	0.63	0.67	0.58	0.65	3.02	2.87	0.49	0.43	0.51	0.53
	5 min	kettle	229.1	245.0	21.7	12.8	30.2	30.4	0.12	0.09	0.00	0.00	0.00	0.00	11.03	11.79	1.21	0.72	1.69	1.70
		freezer	30.6	138.5	29.5	32.2	29.4	30.2	0.44	0.44	0.46	0.45	0.45	0.45	1.36	6.16	1.52	1.66	1.51	1.55
		washing machine	205.6	264.5	33.2	24.4	47.8	54.9	0.21	0.06	0.36	0.63	0.18	0.10	5.36	6.89	1.00	0.74	1.45	1.66
		computer	40.1	30.8	5.7	4.7	5.2	5.5	1.00	0.85	1.00	1.00	1.00	1.00	2.65	2.04	0.38	0.31	0.35	0.37
		micro-wave	284.7	386.7	13.4	9.4	13.3	15.0	0.04	0.04	0.12	0.30	0.13	0.09	27.93	37.94	1.49	1.04	1.48	1.67
		television	43.0	70.6	10.5	9.0	14.7	14.8	0.48	0.3	0.56	0.68	0.53	0.56	2.64	1.61	0.44	0.38	0.62	0.62
	15 min	kettle	204.6	195.8	24.8	28.6	32.6	31.7	0.11	0.16	0.00	0.00	0.00	0.00	9.86	9.44	1.39	1.61	1.82	1.78
		freezer	29.4	35.9	26.7	25.3	26.0	24.8	0.39	0.51	0.55	0.55	0.55	0.55	1.31	1.60	1.37	1.30	1.34	1.28
		washing machine	288.7	312.6	43.3	39.9	45.5	52.1	0.17	0.12	0.27	0.35	0.29	0.21	7.53	8.15	1.31	1.21	1.38	1.58
		computer	22.3	38.5	5.8	6.3	5.7	5.7	1.00	0.82	1.00	1.00	1.00	1.00	1.47	2.55	0.39	0.42	0.38	0.38
		micro-wave	170.6	257.4	12.3	11.5	15.1	14.5	0.07	0.07	0.28	0.27	0.15	0.15	16.78	25.32	1.36	1.28	1.68	1.61
		television	58.8	109.4	12.6	13.6	15.2	18.3	0.47	0.4	0.53	0.42	0.47	0.53	2.2	4.08	0.53	0.57	0.64	0.77
ECO	1 min	kettle	69.1	56.2	25.0	21.8	24.8	29.6	0.09	0.06	0.00	0.00	0.00	0.00	16.06	13.06	7.48	6.52	7.42	8.86
		freezer	44.0	64.8	13.1	18.0	15.2	14.5	0.77	0.40	0.83	0.82	0.75	0.77	2.34	3.44	0.69	0.94	0.80	0.76
		washing machine	79.8	114.1	27.7	33.1	61.5	51.0	0.16	0.19	0.23	0.20	0.19	0.21	3.47	4.96	1.23	1.46	2.72	2.26
		computer	41.0	32.3	17.5	19.1	19.1	17.0	0.49	0.43	0.45	0.45	0.45	0.45	2.06	1.62	0.94	1.02	1.02	0.91
		micro-wave	419.9	551.5	13.7	11.5	9.5	15.8	0.01	0.01	0.03	0.15	0.06	0.04	46.79	61.45	1.57	1.32	1.09	1.81
		television	73.5	93.1	40.6	44.0	35.9	41.5	0.4	0.34	0.38	0.42	0.44	0.38	2.23	2.83	1.28	1.38	1.13	1.31
	5 min	kettle	57.4	57.2	21.5	20.1	32.0	27.8	0.10	0.08	0.00	0.00	0.00	0.00	12.79	12.75	6.27	5.88	9.34	8.12
		freezer	17.7	71.6	13.3	14.3	13.3	13.2	0.76	0.56	0.78	0.76	0.76	0.76	0.94	3.81	0.70	0.76	0.70	0.70
		washing machine	76.9	77.9	48.5	40.8	44.0	45.1	0.14	0.30	0.22	0.19	0.20	0.21	3.36	3.40	2.15	1.81	1.95	2.00
		computer	48.2	53.2	17.0	17.3	16.2	16.0	0.49	0.45	0.44	0.44	0.44	0.44	2.43	2.69	0.91	0.93	0.87	0.86
		micro-wave	296.7	534.6	12.1	9.4	9.5	13.4	0.01	0.01	0.07	0.09	0.04	0.02	33.11	59.66	1.39	1.07	1.08	1.53
		television	76.5	89.4	34.6	34.5	40.0	48.0	0.38	0.35	0.54	0.52	0.35	0.35	2.32	2.71	1.09	1.09	1.26	1.51
	15 min	kettle	75.7	67.4	28.0	23.4	24.1	24.2	0.13	0.11	0.00	0.00	0.00	0.00	17.21	15.33	8.27	6.90	7.10	7.14
		freezer	19.9	29.1	10.1	10.0	10.3	10.3	0.77	0.74	0.84	0.84	0.84	0.84	1.06	1.55	0.53	0.53	0.54	0.54
		washing machine	87.4	102.2	40.9	43.6	40.3	49.7	0.17	0.30	0.17	0.21	0.22	0.25	3.85	4.50	1.81	1.93	1.78	2.20
		computer	42.5	43.9	15.9	15.9	16.1	16.0	0.50	0.46	0.45	0.45	0.45	0.45	2.15	2.22	1.81	1.93	1.78	2.20
		micro-wave	167.7	341.2	8.4	7.3	10.7	11.0	0.03	0.03	0.05	0.00	0.04	0.02	18.79	38.23	0.96	0.83	1.23	1.26
		television	80.5	85.6	41.1	42.9	46.7	50.3	0.36	0.36	0.35	0.26	0.32	0.34	2.44	2.6	1.29	1.35	1.47	1.58

06/03/2015, and were tested on data from the subsequent month, spanning 07/03/2015 to 06/04/2015. For cross-dataset validation, models trained on the REFIT interval were tested against ECO dataset measurements, specifically from 02/06/2012 to 01/07/2012. This cross-dataset approach was employed to ascertain the generalizability of the NILM algorithms across different data sources. A selection was made for REFIT: Building 8 and ECO: Buildings 1, 2 and 5 due to the presence of a greater number of common appliances, which facilitated a

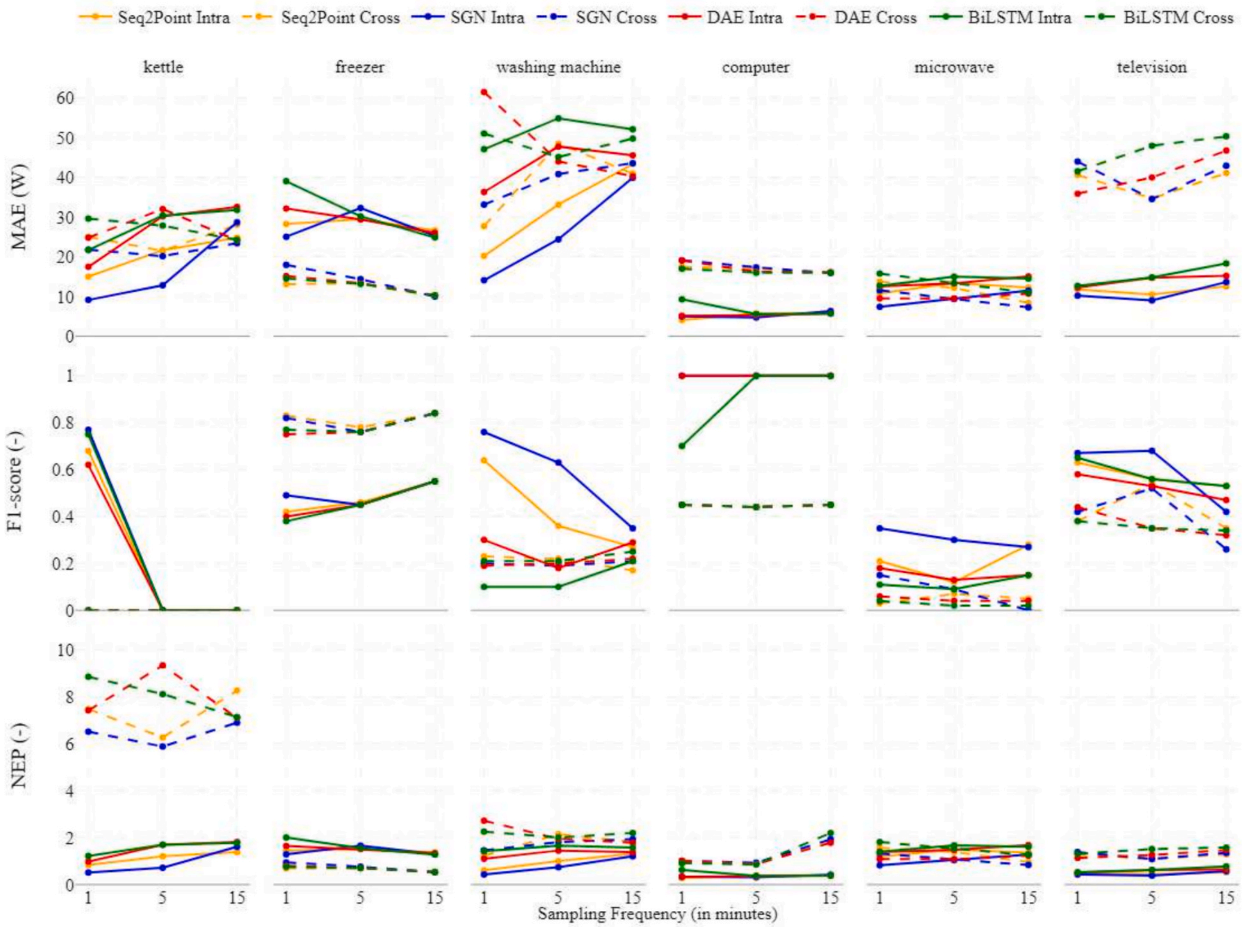

Fig. 5. Variation in the deep learning based NILM algorithm performances over different sampling frequency.

consistent comparative analysis. The appliances analysed included freezer, kettle, computer, washing machine, microwave and television. Additionally, to investigate the influence of data granularity on the accuracy of disaggregation, all datasets were down sampled to the desired sampling periods, i.e. 1, 5, and 15 min. Down sampling was performed utilizing functions provided by the nilmtk framework, ensuring uniformity in the data preparation process across all experiments. Additionally, comparison of the best performing algorithm's (SGN) performance in disaggregating computer's usage with and without washing machine usage in the background was also investigated. The study examines two key scenarios: Computer without Washing machine (CoWo) and Computer with Washing machine (CaWa), tested at three different sampling frequencies with the REFIT: building 8 data. For each of the cases, 3 time periods were selected accordingly (CoWo-1: Nov 16–19, 2014; CoWo-2: Sep 15–17, 2014; CoWo-3: Jul 29-Aug 2, 2014; CaWa-1: Sep 3–6, 2014; CaWa-2: Oct 14–18, 2014; CaWa-3: Aug 28–30, 2014). This analysis aimed to evaluate the performance of NILM algorithms in accurately detecting and disaggregating individual appliance usage patterns under both isolated and concurrent operational scenarios.

3. Results

This chapter presents the performance of six different algorithms – FHMM, CO, Seq2Point, SGN, DAE, and BiLSTM – across various appliances and at different sampling periods. The evaluation of NILM algorithms requires rigorous testing across different scenarios to assess their real-world applicability and generalization capabilities. This analysis presents a comprehensive assessment through both intra-dataset and cross-dataset validation approaches. Intra-dataset validation (REFIT-to-REFIT) examines how well algorithms perform when trained and tested on data from the same building environment, representing ideal conditions where the model learns and operates within a consistent context. Cross-dataset validation (REFIT-to-ECO) tests the algorithms' ability to generalize across different building environments and usage patterns, simulating real-world deployment challenges where algorithms must perform reliably in previously unseen settings. This distinction is crucial as it highlights not only the algorithms' performance under controlled conditions but also their practical applicability across diverse residential settings. Table 2 and Fig. 5 show the performance of NILM algorithms for various appliances at different sampling frequencies in terms of MAE, F1 Score and NEP. The table encompasses experimentation with both intra dataset and cross dataset validation which has been discussed in detail in the following sub-sections.

Intra dataset Performance (REFIT-to-REFIT): The intradataset evaluation, where both training and testing were conducted on the REFIT dataset, showed the following trends:

a) Dynamic Appliances: For the kettle, SGN demonstrated exceptional accuracy, achieving a MAE of 9.11 W, an F1 score of 0.77, and a NEP of 0.51 at 1-minute sampling. Seq2Point followed closely with comparable F1 scores and NEP values, further showcasing its ability to model the sharp transitions in the kettle's usage. In contrast, FHMM and CO struggled to capture the kettle's dynamic patterns. FHMM recorded a MAE of 173.97 W and an F1 score of 0.15, while CO had the lowest performance with a MAE of 293.98 W and F1 score of 0.07. For the microwave, SGN once again led the results, achieving a MAE of 7.37 W and a low NEP of 0.82 at 1-minute sampling, highlighting its ability to identify the microwave's distinct power spikes. Seq2Point followed closely with MAE of 10.78 W. FHMM and CO showed significant limitations, with MAEs of 237.97 W and 448.45 W, respectively. Their F1 scores of 0.02 indicate poor state detection accuracy.

b) Steady-State Appliances: In case of freezer, SGN achieved the best performance at 1-minute sampling, recording a MAE of 25.03 W, an F1 score of 0.50, and an NEP of 1.29, highlighting its precision even for steady-state appliances. FHMM performed relatively well for this appliance, with MAE of 48.02 W and an F1 score of 0.42, while CO struggled, recording a MAE of 139.95 W. For the computer, Seq2-Point achieved the best performance at 1-minute sampling, achieving a MAE of 4.03 W, an F1 score of 1, and an NEP of 0.27, closely followed by SGN. FHMM and CO performed decently compared to its performance on other appliances, with an F1 score of 1 and 0.68 respectively.

c) Mixed Behaviour Appliances: For washing machine, SGN was yet again the best performing algorithm at 1-minute sampling. It obtained a MAE of 14.06 W, an F1 score of 0.76 and a low NEP of 0.43. Similar performance was also obtained with Seq2Point only slightly worse. The benchmark algorithms performed significantly worse with MAEs above 224 W, F1 scores below 0.17. The results obtained for the television was slightly different. SGN was the best performing algorithm with MAE of 9.01 W, F1 Score of 0.68 and NEP of 0.38. However, it is interesting to note that this best performance was obtained with 5-minutes sampling frequency. FHMM and CO provided unsatisfactory results with MAEs of 81.15 W and 76.66 W respectively.

d) Impact of Sampling Frequency: At higher sampling frequencies (e.g., 1 min), the algorithms perform better overall due to finer resolution of data capturing the transitions of appliances. As the sampling frequency decreases to 5 min or 15 min, there is a notable increase in MAE and a reduction in F1 scores, particularly for high-dynamic appliances like the kettle and microwave. At 1 min sampling frequency the F1 score using SGN for kettles was 0.77 which drops to 0 at 15 min sampling frequency. In case of steady state appliances, the effect is less pronounced. In fact, for computer there is no change in F1 score and only a slight change in MAE values between 1- and 15-minutes sampling frequency.

Cross-Dataset Performance (REFIT-to-ECO): The cross-dataset evaluation, where training was conducted on REFIT and testing on ECO, revealed a noticeable performance drop for all algorithms.

a) Dynamic Appliances: For kettle, all the algorithms performed quite poorly with the maximum F1 score of 0.13. The lowest MAE value of 20.14 W was recorded with SGN at 5-minutes sampling frequency. However, an F1 Score of 0 in the same configuration indicates a complete failure in detecting kettle activation states. In case of microwave, the performance of both the deep-learning and benchmark algorithms are not much better. SGN performs the best with MAEs of 7.25 W and 9.35 W at 15-minutes and 5-minutes sampling frequency. However, low (0, 0.1) F1 scores illustrate failure in appliance state detection.

b) Steady-State Appliances: Freezer and computer showed better generalizability across datasets compared to dynamic appliances. In case of freezer, SGN recorded the best values with MAE of 9.99 W, an F1 score of 0.84, and an NEP of 0.53, demonstrating relatively consistent performance. FHMM performed comparably, achieving an F1 score of 0.77, suggesting that steady-state appliances are less sensitive to cross-dataset differences. For computer, SGN and BiLSTM performed the best with MAE of 16 W, an F1 score of 0.45, and an NEP of 0.86, showing reasonable generalization despite cross-dataset challenges. FHMM and CO showed reduced but fair performance, with F1 scores around 0.45, highlighting their ability to generalize better for steady-state appliances.

c) Mixed Behaviour Appliances: For washing machine, Seq2Point recorded a MAE of 27.71 W and a NEP of 1.23 and was the best performing algorithm. FHMM and CO struggled with generalization, recording MAEs of 79.83 W (FHMM) and 114.08 W (CO), and very low F1 scores (0.16 and 0.19, respectively) at 1-minute sampling frequency. In case of television, DAE, SGN and Seq2Point perform the best at 1-,5- and 15-minutes sampling frequency with MAEs ranging between 35.9–41.1 W and F1 Scores in the range of 0.4–0.5. As expected, the benchmark algorithms fare worse but provide fair results with F1 scores on average between 0.35–0.4.

d) Impact of Sampling Frequency: Performance degradation was more pronounced at lower sampling frequencies in the cross-dataset setting. For example, Seq2Point's F1 score for the washing machine dropped from 0.27 (REFIT-to-REFIT) to 0.17 (REFIT-to-ECO) at 15-minute sampling. Seq2Point and SGN remained relatively robust at lower sampling frequencies, while FHMM and CO suffered substantial drops in accuracy.

4. Discussion

The evaluation of intradataset and cross-dataset performance highlighted key insights into the robustness and adaptability of the NILM

Table 3
Composite scores of the different NILM algorithms at different sampling frequencies.

Sample Freq.	Appliance	REFIT-to-REFIT						REFIT-to-ECO					
		FHMM	CO	Seq2Point	SGN	DAE	BiLSTM	FHMM	CO	Seq2Point	SGN	DAE	BiLSTM
1 min	kettle	2.01	2.93	0.36	0.23	0.45	0.35	2.91	2.35	1.17	1.00	1.16	1.41
	freezer	0.96	2.57	0.64	0.51	0.74	0.90	1.43	2.60	0.17	0.37	0.33	0.28
	washing machine	2.83	2.39	0.41	0.24	0.87	1.15	2.04	2.81	0.77	0.93	1.60	1.34
	computer	2.00	1.68	0.00	0.05	0.06	0.57	2.51	1.82	0.60	0.74	0.74	0.55
	microwave	2.02	2.98	0.81	0.65	0.84	0.92	2.50	2.99	0.99	0.86	0.94	0.98
	television	2.69	2.51	0.42	0.33	0.48	0.42	1.90	2.66	0.79	0.87	0.56	0.82
5 min	kettle	2.74	2.91	1.08	1.00	1.16	1.16	2.90	2.91	1.09	1.00	1.82	1.53
	freezer	0.57	2.56	0.57	0.64	0.58	0.60	0.40	2.44	0.23	0.28	0.24	0.24
	washing machine	2.30	2.94	0.72	0.37	1.03	1.18	2.81	2.70	1.20	0.81	0.98	1.03
	computer	2.00	1.63	0.06	0.00	0.03	0.05	2.23	2.55	0.62	0.64	0.58	0.56
	microwave	2.42	2.96	0.90	0.70	0.89	0.94	2.08	2.99	0.94	0.91	0.96	1.00
	television	2.07	2.24	0.49	0.32	0.67	0.64	2.14	2.65	0.46	0.48	0.85	1.15
15 min	kettle	2.89	2.74	1.00	1.05	1.09	1.08	2.87	2.55	1.22	1.00	1.03	1.04
	freezer	1.12	2.49	0.90	0.56	0.74	0.45	1.27	2.26	0.17	0.16	0.19	0.19
	washing machine	2.65	2.88	0.76	0.65	0.76	0.89	2.35	2.70	0.85	0.89	0.78	1.06
	computer	1.01	2.18	0.01	0.04	0.00	0.00	2.29	2.54	0.62	0.89	0.56	1.51
	microwave	2.22	2.93	0.73	0.73	0.88	0.88	1.93	2.97	0.96	1.00	0.98	1.00
	television	1.48	2.60	0.47	0.60	0.59	0.60	2.40	2.64	0.65	0.83	0.94	1.09

Table 4
Best performing algorithms for various appliances at different sampling frequencies.

REFIT-to-REFIT																		
Sample Freq.	1 min						5 min						15 min					
Appliance	kettle	freezer	washing machine	computer	microwave	television	kettle	freezer	washing machine	computer	microwave	television	kettle	freezer	washing machine	computer	microwave	television
Algorithm	SGN	SGN	SGN	Seq2Point	SGN	SGN	SQN	Seq2Point	SGN	SGN	SGN	SGN	Seq2Point	SGN	SGN	DAE	SGN	Seq2Point
REFIT-to-ECO																		
Sample Freq.	1 min						5 min						15 min					
Appliance	kettle	freezer	washing machine	computer	microwave	television	kettle	freezer	washing machine	computer	microwave	television	kettle	freezer	washing machine	computer	microwave	television
Algorithm	SGN	Seq2Point	Seq2Point	BiLSTM	SGN	DAE	SGN	Seq2Point	Seq2Point	BiLSTM	SGN	Seq2Point	SGN	SGN	Seq2Point	DAE	Seq2Point	Seq2Point

algorithms. Deep learning models demonstrated strong performance when training and testing were conducted on the same dataset. SGN and Seq2Point consistently outperformed other deep learning and benchmark algorithms across all appliances and sampling frequencies, leveraging their ability to model appliance-specific patterns effectively. For instance, Seq2Point's F1 score of 0.93 for the microwave at 1-minute sampling showcases its ability to detect state transitions accurately in intradataset evaluations. Benchmark algorithms like FHMM and CO performed relatively better for steady-state appliances, such as the freezer, but struggled with dynamic appliances due to their limited capacity to model transient power usage.

Cross-dataset evaluations revealed a clear degradation in performance for all algorithms, reflecting the domain gap between datasets. Seq2Point and SGN remained the most robust but experienced reduced accuracy due to variations in appliance usage patterns between REFIT and ECO. For example, Seq2Point's MAE for the kettle increased from 14.99 W in REFIT-to-REFIT to 25 W in REFIT-to-ECO, and its F1 score dropped from 0.93 to 0.68. FHMM and CO struggled significantly in the cross-dataset setting, as their statistical nature limited their ability to generalize across datasets. For dynamic appliances like microwaves, FHMM recorded a MAE of 419.93 W in REFIT-to-ECO, compared to 237.97 W in REFIT-to-REFIT.

To determine the best-performing algorithm for each sampling frequency and appliance, a systematic and fair evaluation approach was implemented. First, the metrics with differing scales, i.e. MAE and NEP were normalized using min–max normalization. This transformation scaled the values to a range between 0 and 1, ensuring all metrics including F1 Score were on the same order of magnitude and directly comparable. After normalization, a multi-metric scoring approach was employed. Equal weights were assigned to each normalized metric, reflecting the assumption that all metrics are equally important for the task. The overall performance score for each algorithm was computed as the weighted sum of the normalized values, aggregating the results of MAE, F1 Score and NEP. The equation 1 shows the mathematical formula that was used to calculate a single representative score from the 3 metrics.

$$Composite\,Score = norm.\,MAE + (1 - F1\,Score) + norm.\,NEP$$

This composite score allowed for an objective comparison of algorithms across multiple dimensions of performance. Table 3 shows the calculated composite scores of the different NILM algorithms using Eq. (1). This process is critical because it prevents dominance by any single metric and ensures that the selected algorithm performs consistently well across all evaluation criteria. Additionally, it provides a structured and transparent methodology for algorithm selection, which is particularly important when working with diverse datasets and performance measures. Table 4 below shows the best performing algorithms for various appliances at different sampling frequencies obtained from the composite score.

In general, across both intra dataset and cross dataset validation, SGN and Seq2Point excelled in handling transient behaviour, as seen in the kettle and microwave results. Their high F1 scores and low NEP values highlight their ability to detect precise state transitions and maintain prediction accuracy, even at coarser resolutions. Mixed-behaviour appliances, such as TVs and washing machines, presented a balanced challenge between dynamic and steady-state characteristics. As seen in Table 3, the deep learning-based algorithms especially SGN and Seq2Point successfully captured these nuances, while FHMM and CO lacked the robustness to handle the variability, leading to higher error metrics. While benchmark algorithms performed relatively better for appliances like freezers, their results were still inferior to Seq2Point and SGN. The ability of deep learning models to account for subtle variations in steady-state appliances contributed to their superior NEP and MAE values. Lower sampling frequencies also amplified the performance gap between intradataset and cross-dataset evaluations. Seq2Point and SGN exhibited better adaptability, maintaining

Table 5
Comparison of SGN performance for computer usage disaggregation with and without washing machine usage.

Case	CoWo-1			CoWo-2			CoWo-3			CaWa-1			CaWa-2			CaWa-3		
Freq. (min)	1	5	15	1	5	15	1	5	15	1	5	15	1	5	15	1	5	15
MAE (W)	16.9	20.4	22.0	13.8	19.1	19.5	5.3	15.8	15.8	11.6	22.3	20.5	10.7	19.7	20.5	9.8	19.1	20.4
F1 Score (-)	0.70	0.36	0.36	0.67	0.37	0.38	0.75	0.22	0.24	0.65	0.32	0.33	0.71	0.29	0.29	0.66	0.32	0.31
NEP (-)	0.66	0.79	0.85	0.55	0.76	0.77	0.36	1.07	1.07	0.46	0.88	0.81	0.51	0.95	0.98	0.38	0.75	0.80

competitive F1 scores for dynamic appliances at lower resolutions, whereas FHMM and CO suffered disproportionately. Multivariate event detection methods can also significantly improve performance and accuracy in non-intrusive load monitoring for smart homes and residential buildings compared to classical scalar ones [46]. Furthermore, using harmonic current vectors in NILM algorithms significantly enhances disaggregation performance, making them particularly suitable for smart meters [47]. The ECO dataset appears to present additional challenges due to differences in appliance usage patterns and noise levels. This disparity highlights the importance of training algorithms on diverse datasets to improve generalization.

The Table 5 presents comparison of SGN performance for computer usage disaggregation with and without washing machine usage. MAE values show an interesting pattern where CaWa scenarios consistently demonstrate lower errors at the 1-minute sampling rate (9.78–11.57 W) compared to CoWo scenarios (13.81–16.89 W). This suggests the algorithm might perform better at estimating power consumption when both appliances operate concurrently at higher sampling frequencies. Regarding F1 Scores, both scenario types show comparable detection capabilities, generally ranging between 0.65–0.75 at 1-minute sampling. This indicates the algorithm's ability to correctly identify device states remains relatively consistent whether the washing machine is operating or not. NEP values reveal that CoWo scenarios show greater variability across different test cases. Most notably, CoWo-3 demonstrates excellent performance at 1-minute sampling (0.36) but deteriorates dramatically at lower frequencies (exceeding 1.0). Meanwhile, CaWa scenarios maintain more consistent NEP values, suggesting more predictable estimation errors when both appliances operate together. Contrary to what might be expected, the algorithm doesn't necessarily perform worse when detecting computer usage with a concurrent washing machine operation. The washing machine's distinct power signature may help the algorithm's overall detection capabilities in some instances, particularly at higher sampling frequencies. However, more robust cases with other appliances and buildings' data needs to be further investigated to make a more conclusive statement on the above.

5. Conclusion

In conclusion, the performance of NILM algorithms is highly dependent on the appliance, sample period, and dataset specificity. In general, SGN emerges as the most versatile algorithm, excelling in capturing both dynamic (e.g., kettle, microwave) and steady-state (e.g., freezer, computer) appliance behaviours across different sampling rates and validation scenarios. Seq2Point is a strong alternative, especially for steady-state and mixed-behaviour appliances at higher frequencies. Cross-dataset performance highlights the need for fine-tuning or transfer learning, with SGN and BiLSTM showing relatively better generalization capabilities. The significant drop in performance when models trained on REFIT are tested on ECO, points to the importance of dataset diversity in training for better generalizability. This analysis underscores the need to design NILM algorithms that are robust across different sampling rates and capable of transferring knowledge across diverse datasets to ensure practical applicability in real-world scenarios. It is crucial also to consider the impact of geographical and cultural differences. The datasets originate from different countries which could introduce a variety of discrepancies. These may include differences in appliance specifications due to varying manufacturing standards, energy consumption patterns influenced by local climate, and usage behaviours shaped by cultural habits. All these factors can contribute to the observed drop in performance and underscore the importance of region-specific training data to achieve optimal model generalization.

Considering the observed performance variations when generalizing models from one dataset to another, incorporating region-specific information into algorithms could significantly enhance their performance. In future studies the author also intends to explore the potential of super-resolution methodologies, which aim to reconstruct high-resolution data from low-resolution measurements. Based on preliminary research, these methodologies can potentially improve the performance of deep learning based NILM algorithms. These techniques comprise of both non-machine learning (such as interpolation, signal processing, and statistical methods) and machine learning-based methods (such as Convolutional Neural Networks and Generative Adversarial Networks), and hold the promise for enhancing data completeness, capturing intricate energy consumption patterns, and ultimately improving the performance of NILM systems. Our findings endorse the pursuit of NILM algorithms that leverage super-resolution to achieve robustness across various sampling rates and to fortify knowledge transfer capabilities, ensuring their efficacy and practical applicability in diverse real-world energy monitoring applications.

CRediT authorship contribution statement

Arnab Chatterjee: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Philipp Heer:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Arnab Chatterjee reports financial support was provided by Swiss Federal Office of Energy. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Code repository

https://github.com/Vincisteel/NILM_Experimentation.git.

Data availability

Data will be made available on request.

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